ETF Risk-Reward Tradeoff Analysis:
Evaluating 2,259 ETFs Using ETFG Red Diamond and Green Diamond Scoring Metrics

Shubham Oza
MS Financial Technology and Analytics
Naveen Jindal School of Management
University of Texas at Dallas

March 2026

Abstract

This study examines the risk-reward tradeoff across exchange-traded funds (ETFs) using the ETFG Analytics dataset sourced from the Wharton Research Data Services (WRDS) platform, covering January 2, 2024 through March 2, 2026. The dataset comprises 1,224,378 daily ETF observations across 2,259 distinct funds - one of the largest ETF panel datasets employed in academic study of this type. Leveraging the ETFG proprietary Red Diamond (risk) and Green Diamond (reward) scoring framework, this analysis finds a stark and systematic imbalance: the average ETF carries a risk score of 7.17 out of 10 while delivering a reward score of only 2.79 out of 10, yielding a mean reward-to-risk (RR) ratio of 0.39. Over 56% of ETFs fall in the High or Very High risk tier, and 25% score exactly zero on reward. These findings challenge the assumption that the ETF market efficiently prices risk and provide a practical framework for risk-aware ETF selection using the RR ratio and quadrant segmentation methodology.

1. Background

Exchange-traded funds have fundamentally transformed the investment landscape since their introduction in the early 1990s. As of 2024, the global ETF industry manages over $10 trillion in assets, with U.S.-listed ETFs accounting for the majority of that figure (BlackRock, 2024). The proliferation of ETF products - now exceeding 8,000 U.S.-listed funds - has expanded investor access to diversified, low-cost exposure across asset classes, geographies, sectors, and thematic strategies. However, this democratization of investing has also introduced complexity: not all ETFs are created equal, and the risk-reward characteristics of individual funds can vary substantially even within the same category.

Traditional risk assessment frameworks for ETFs typically focus on standard deviation, beta, and tracking error as primary risk metrics. While these measures capture meaningful dimensions of risk, they fail to account for structural risks unique to ETF products - such as fund complexity, counterparty exposure in synthetic ETFs, liquidity mismatch between fund holdings and secondary market trading, and geographic concentration. The ETFG (Exchange Traded Funds Global) scoring methodology addresses this gap through its proprietary Red Diamond risk scoring and Green Diamond reward scoring systems, which aggregate multiple risk and reward dimensions into composite scores specifically designed for ETF evaluation.

The present study is motivated by a critical gap: while ETFG scores are used by institutional practitioners, no academic study has systematically examined the distribution of these scores across the full ETF universe over a multi-year panel. By analyzing 1,224,378 daily observations across 2,259 ETFs from January 2024 through March 2026, this paper provides the first large-scale empirical characterization of the ETFG risk-reward landscape and its implications for portfolio construction.

1.1 Literature Review

Prior research on ETF risk and performance has identified several key findings that contextualize this study. Agapova (2011) demonstrated that ETFs and mutual funds differ significantly in their liquidity provision, with ETFs exhibiting tighter bid-ask spreads but being more susceptible to intraday volatility shocks. Ben-David, Franzoni, and Moussawi (2018) found that ETF ownership

increases the co-movement of underlying securities, amplifying systemic risk particularly during periods of market stress - a finding consistent with the elevated risk scores observed in this study.

Research on risk-adjusted performance has shown mixed results. Rompotis (2009) found that ETFs generally underperform their benchmarks on a risk-adjusted basis after accounting for trading costs, while other studies have documented superior performance in less liquid market segments. More recently, Israeli, Lee, and Sridharan (2017) documented that increased ETF ownership is associated with reduced price informativeness of individual securities, raising broader market efficiency concerns relevant to the reward scoring framework.

Within the practitioner literature, ETFG's proprietary scoring methodology has been employed by institutional investors to screen and rank ETFs across risk dimensions. However, to the author's knowledge, no published academic study has characterized the full distributional properties of ETFG scores across the complete U.S. ETF universe over a multi-year period. This study fills that gap.

2. Definition of the Problem

The central problem motivating this analysis is the challenge of evaluating ETFs on a comparable, multi-dimensional risk-reward basis. While numerous metrics exist for assessing individual dimensions of ETF performance, investors lack a unified framework for understanding how risk and reward interact at the fund level and which fund characteristics are most associated with risk-adjusted attractiveness.

Preliminary examination of the ETFG data reveals a troubling pattern: with a mean risk score of 7.17 and a mean reward score of 2.79 across 2,259 ETFs, the average ETF in the universe is a poor risk-reward proposition. The interquartile range of reward scores (0.00 to 4.82) further reveals that at least 25% of ETFs score zero on reward - providing no measurable investment attractiveness while still carrying non-trivial risk. This distributional asymmetry raises fundamental questions about how well the ETF market compensates investors for the risks they bear.

This study seeks to answer four primary research questions:

- Is there a systematic positive relationship between ETFG risk scores (Red Diamond) and reward scores (Green Diamond) across the 2,259-ETF universe?

- Which risk components (volatility, deviation, country, structure, liquidity, efficiency) contribute most to total risk scores, and how are these distributed across the ETF universe?
- Can ETFs be meaningfully segmented into distinct risk-reward quadrants, and which fund types populate the optimal High Reward / Low Risk quadrant?
- Which ETFs offer the most attractive risk-adjusted profiles as measured by the reward-to-risk ratio, and what characteristics define these funds?

Answering these questions has direct practical relevance for portfolio construction, ETF screening, and risk management. The analysis is particularly timely given the market volatility experienced in 2024-2025, which stressed ETF structures in ways not seen in prior low-volatility periods.

3. Explanation of the Data

3.1 Data Source

The dataset was obtained from the Wharton Research Data Services (WRDS) platform, specifically from the ETFG Analytics library (etfg.analytics). WRDS is a leading academic data platform used by researchers at over 450 institutions globally. ETFG (Exchange Traded Funds Global) is a specialist ETF data and analytics provider that computes daily risk and reward scores for U.S.-listed ETFs using a proprietary methodology. Data was accessed via the WRDS Python API using direct SQL query against the etfg.analytics table.

3.2 Dataset Characteristics

The dataset spans **January 2, 2024 through March 2, 2026** - a 26-month panel covering 489 trading days. The full dataset contains **1,224,378 observations** across **2,259 distinct ETF tickers**, averaging approximately 542 daily observations per ETF. The dataset was retrieved with all 34 available variables, of which 10 core variables were retained for analysis (see Exhibit 1).

Exhibit 1 summarizes the key variables used in this analysis:

Exhibit 1: Key Variables from ETFG Analytics Dataset

Variable	Label	Description
as_of_date	Trade Date	Primary time index; daily frequency
composite_ticker	ETF Ticker Symbol	Unique ETF identifier; panel key
risk_total_score	Red Diamond Risk Score	Composite 0–10 risk score (higher = riskier)
risk_volatility	Volatility Component	Standard deviation & beta exposure sub-score
risk_deviation	Deviation Component	Tracking error relative to benchmark
risk_country	Country Component	Geographic concentration risk
risk_structure	Structure Component	Fund structure & counterparty risk
risk_liquidity	Liquidity Component	Bid-ask spread & volume sub-score
risk_efficiency	Efficiency Component	Expense ratio & operational risk
reward_score	Green Diamond Reward Score	Composite 0–10 reward/attractiveness score

3.3 Descriptive Statistics

Exhibit 2 presents the full descriptive statistics for the two primary outcome variables - the Red Diamond risk score and the Green Diamond reward score - across all 1,224,378 observations.

Exhibit 2: Descriptive Statistics - Risk and Reward Scores

Statistic	Risk Total Score	Reward Score
Count	1,224,378	1,224,378
Mean	7.17	2.79
Std Dev	2.43	2.68
Min	1.00	0.00
25th Pct	4.94	0.00
Median	7.63	3.06
75th Pct	9.50	4.82
Max	10.00	10.00

The descriptive statistics reveal a striking asymmetry in the risk-reward profiles of ETFs. The mean risk score of 7.17 (on a 0–10 scale) is substantially elevated, with a median of 7.63 indicating

that more than half of all ETF-day observations carry a risk score above 7.6. In contrast, the mean reward score of 2.79 is well below the midpoint of the scale, with a 25th percentile of exactly 0.00 - meaning at least one quarter of all ETF-day observations carry zero reward. The resulting mean RR ratio of 0.39 (2.79 ÷ 7.17) indicates that the typical ETF delivers less than 40 cents of reward for every unit of risk carried.

3.4 Risk Tier Segmentation

To facilitate cross-sectional analysis, ETFs were classified into four risk tiers based on their risk_total_score values. Exhibit 3 presents these tiers along with their approximate prevalence in the dataset and typical fund types:

Exhibit 3: ETF Risk Tier Classification

Tier	Score Range	% of ETFs	Typical Fund Types
Low	0.0 – 3.0	~5%	Broad US equity index; large-cap domestic ETFs
Medium	3.0 – 5.0	~13%	Diversified international and mixed asset ETFs
High	5.0 – 7.0	~25%	Sector, thematic, and multi-asset ETFs
Very High	7.0 – 10.0	~57%	Leveraged, single-country, and speculative ETFs

The risk tier distribution confirms that the ETF universe is heavily skewed toward higher-risk funds. Approximately 57% of ETF-day observations fall in the Very High tier (score 7.0–10.0), consistent with the median risk score of 7.63. Only about 5% of observations fall in the Low risk tier, indicating that truly low-risk ETFs represent a small fraction of the available universe.

3.5 Data Cleaning and Preparation

The raw dataset was processed through the following steps prior to analysis:

- Date Formatting: The as_of_date field was parsed as a date type and used as the primary temporal index.
- Missing Values: Observations with null values in risk_total_score or reward_score were excluded. The dataset contained no null values in these fields across the 1,224,378 rows.
- Derived Metrics: A reward-to-risk ratio (rr_ratio) was computed as reward_score divided by risk_total_score, handling the rare case of zero risk scores to avoid division errors.

- Risk Tier Classification: Observations were binned into four risk tiers (Low, Medium, High, Very High) using fixed score thresholds.
- Quadrant Classification: Each observation was assigned to one of four quadrants based on whether its risk and reward scores fell above or below 5.0 (the midpoint of the 0–10 scale).
- Year-Month Field: A year_month string field (e.g., '2024-01') was derived for time-series aggregation.

4. Analysis

4.1 Methodology

The analysis employs a combination of descriptive statistics, distributional analysis, correlation analysis, time-series trend analysis, and quadrant-based fund segmentation. All data processing was performed using Python (pandas, numpy) with the WRDS Python API for data retrieval. Interactive visualization and exploratory analysis were conducted using Microsoft Power BI Desktop, with data transformations implemented in Power Query M.

The analytical framework proceeds in four stages: (1) characterization of the risk and reward score distributions and their implications; (2) risk tier analysis examining the prevalence and composition of ETFs across the four risk tiers; (3) quadrant segmentation identifying the four risk-reward combinations and their population; and (4) RR ratio analysis identifying the most attractive ETFs on a risk-adjusted basis.

4.2 Risk-Reward Distributional Findings

The most significant finding of this analysis is the systematic imbalance between risk and reward scores across the ETF universe. As documented in Exhibit 2, the mean risk score (7.17) is more than 2.5 times the mean reward score (2.79). This divergence is not merely a statistical artifact - it reflects a genuine structural feature of the ETF universe in which risk is widespread and reward is concentrated.

The distribution of risk scores is left-skewed, with a long tail at the low end and the bulk of observations concentrated in the 5-10 range. The 75th percentile risk score of 9.50 indicates that three quarters of ETF-day observations carry a risk score below 9.5, but the remaining quarter are

clustered near the maximum, reflecting a concentration of very high-risk products at the top of the distribution.

The reward score distribution tells a different story. With a 25th percentile of exactly 0.00, at least one quarter of ETF-day observations carry zero measured reward. The median reward score of 3.06 confirms that reward is not broadly distributed - rather, it is concentrated in a relatively small subset of ETFs. The standard deviation of reward scores (2.68) is nearly equal to the mean (2.79), indicating extreme dispersion relative to the central tendency. This right-skewed, zero-inflated distribution of reward scores is a defining characteristic of the ETFG universe.

4.3 Correlation Between Risk and Reward

A Pearson correlation analysis between the Red Diamond risk score and the Green Diamond reward score across all 1,224,378 observations yields a coefficient of $r = 0.0082$ ($p < 0.001$). This result is among the most striking empirical findings of this study: despite the near-universal assumption in finance that risk and return are positively related, the ETFG scoring framework reveals that risk and reward are essentially uncorrelated across the ETF universe.

The near-zero correlation carries important implications. It means that bearing more risk in ETF selection does not, on average, translate into higher reward scores. High-risk ETFs are just as likely to carry low reward scores as high ones. This directly challenges the efficient market assumption that the ETF market adequately prices risk and provides theoretical justification for the systematic screening approach developed in this study. Investors who simply accept higher risk in exchange for expected higher reward are not rewarded for doing so in this dataset.

The statistical significance of this near-zero correlation ($p < 0.001$) reflects the enormous sample size of 1,224,378 observations rather than economic significance. The practical implication is clear: risk score alone is not a useful predictor of reward score, and investors must evaluate both dimensions independently.

4.4 Risk Component Analysis

The ETFG Red Diamond total risk score aggregates six component scores. Exhibit 5 presents the mean value of each component across all 1,224,378 observations:

Exhibit 5: Mean Risk Component Scores Across 2,259 ETFs (Jan 2024 - Mar 2026)

Component	Mean Score	Interpretation
risk_volatility	**5.11**	Highest component — standard deviation and beta drive risk most
risk_deviation	**5.00**	Near-midpoint — tracking error widely distributed across ETFs
risk_liquidity	**5.03**	High and time-varying — spikes during market stress periods
risk_efficiency	**4.98**	Near-midpoint — expense ratios and operational risk fairly uniform
risk_structure	**4.00**	Lower average — elevated mainly in leveraged and synthetic ETFs
risk_country	**3.49**	Lowest component — concentrated risk only in single-country ETFs

The component analysis reveals a clear hierarchy of risk drivers. Volatility (5.11) is the highest-scoring component on average, confirming its status as the dominant source of ETF risk across the universe. Liquidity (5.03) and deviation (5.00) are nearly equal in average contribution, both clustering near the midpoint of the scale. Efficiency (4.98) follows closely, reflecting the relatively uniform distribution of expense ratios and operational risk across the ETF universe.

The two lowest-scoring components are structure (4.00) and country (3.49). The below-average structure score reflects the fact that most ETFs employ straightforward physical replication rather than complex synthetic structures — complexity-driven structural risk is concentrated in a minority of leveraged, inverse, and derivative-based products. Country risk (3.49) is the lowest component on average, consistent with the prevalence of broadly diversified domestic and multi-country ETFs in the universe; single-country exposure is the exception rather than the norm.

Notably, the total risk score mean of 7.17 substantially exceeds the simple average of the six component means (approximately 4.60), suggesting that the ETFG aggregation methodology applies non-linear weighting or interaction effects that amplify the composite score beyond a simple average of components. This non-linearity underscores the value of using the composite Red Diamond score rather than individual components for holistic risk assessment.

4.5 Quadrant Segmentation Analysis

To provide an actionable framework for ETF evaluation, funds were classified into four quadrants based on the midpoint (5.0) of the 0-10 scoring scale. Exhibit 4 describes these quadrants and their investment implications:

Exhibit 4: ETF Risk-Reward Quadrant Framework

Quadrant	Label	Typical Composition	Portfolio Implication
High Reward / Low Risk	**Optimal**	Broad US equity, IG fixed income ETFs with large AUM	Core portfolio allocation
High Reward / High Risk	**Aggressive**	High-momentum sector ETFs with elevated structural complexity	Tactical / satellite positions
Low Reward / Low Risk	**Defensive**	Short-duration bond, money market, and stable value ETFs	Capital preservation
Low Reward / High Risk	**Avoid**	Leveraged, inverse, single-country EM ETFs	Systematic exclusion recommended

The quadrant population counts, computed across all 1,224,378 ETF-day observations, reveal a striking concentration in the worst-performing quadrant. The Low Reward / High Risk quadrant accounts for 677,378 observations (55.3% of the total), confirming that more than half of all ETF-day observations represent funds simultaneously carrying above-median risk and below-median reward. The Low Reward / Low Risk quadrant accounts for a further 262,173 observations (21.4%), meaning that 76.7% of the entire ETF universe falls in low-reward territory.

The optimal High Reward / Low Risk quadrant accounts for only 52,620 observations (4.3% of the total) - the smallest of the four quadrants by a substantial margin. This finding reinforces the near-zero correlation result: not only are risk and reward essentially uncorrelated, but genuinely attractive ETFs (high reward, low risk) are exceptionally rare, representing fewer than 1 in 23 ETF-day observations. The High Reward / High Risk quadrant accounts for 232,207 observations (19.0%), representing an aggressive but at least reward-positive category.

The High Reward / Low Risk quadrant is populated primarily by broad U.S. equity index funds, investment-grade bond ETFs, and select multi-asset funds with strong flow momentum and low structural complexity. The Low Reward / High Risk quadrant is disproportionately concentrated among leveraged and inverse ETFs, single-country emerging market funds, and thematic ETFs with narrow mandates.

4.6 Reward-to-Risk Ratio Analysis

The RR ratio, defined as the Green Diamond reward score divided by the Red Diamond risk score, provides a single parsimonious metric for risk-adjusted attractiveness. Across the full 2,259-ETF universe, the mean RR ratio is 0.39, confirming that the average ETF delivers less than 40 cents of reward per unit of risk. Exhibit 5 presents the top 10 ETFs by average RR ratio across the study period:

Exhibit 6: Top 10 ETFs by Average Reward-to-Risk Ratio (Jan 2024 - Mar 2026)

Ticker	Fund Category	Avg RR Ratio
ZJPN	Japan equity - hedged/structured mandate	**2.160**
IDHD	International high dividend - developed markets	**1.669**
IHI	U.S. medical devices sector ETF	**1.639**
XLF	U.S. financial sector - broad, large-cap	**1.602**
PBDM	Developed markets - quality factor tilt	**1.590**
VTWO	U.S. small-cap equity - broad Vanguard index	**1.543**
VONV	U.S. value equity - broad Vanguard index	**1.542**
DHVW	Dividend/value - hedged global mandate	**1.514**
FPE	Preferred securities - investment grade focus	**1.490**
PFXF	Preferred securities - ex-financials	**1.489**

The top-10 ETFs by RR ratio span a range of mandates - from Japan-hedged equity (ZJPN) and U.S. financial sector (XLF) to preferred securities (FPE, PFXF) and small-cap value (VTWO, VONV). While diverse in mandate, these funds share common structural characteristics: defined factor tilts, liquid underlying securities, and relatively transparent index methodologies. Notably, broad-market Vanguard funds (VTWO, VONV) appear prominently, consistent with the finding that simple, large, low-cost ETFs dominate the attractive end of the risk-reward distribution.

The highest RR ratio of 2.160 (ZJPN) reflects a period-specific dynamic in which Japan equity delivered strong momentum-driven reward scores while maintaining moderate structural risk. This underscores that RR ratios can be influenced by cyclical factors, reinforcing the recommendation to monitor these metrics dynamically rather than relying on static rankings.

4.7 Power BI Dashboard Visualization

To support interactive exploration of the findings, a four-page Power BI dashboard was developed using the full 1,224,378-observation dataset. The dashboard enables real-time filtering by date, quadrant, risk tier, and individual ETF ticker. Exhibits 7 through 10 present the four dashboard pages.

Exhibit 7: Power BI Dashboard - Executive Overview (Page 1)

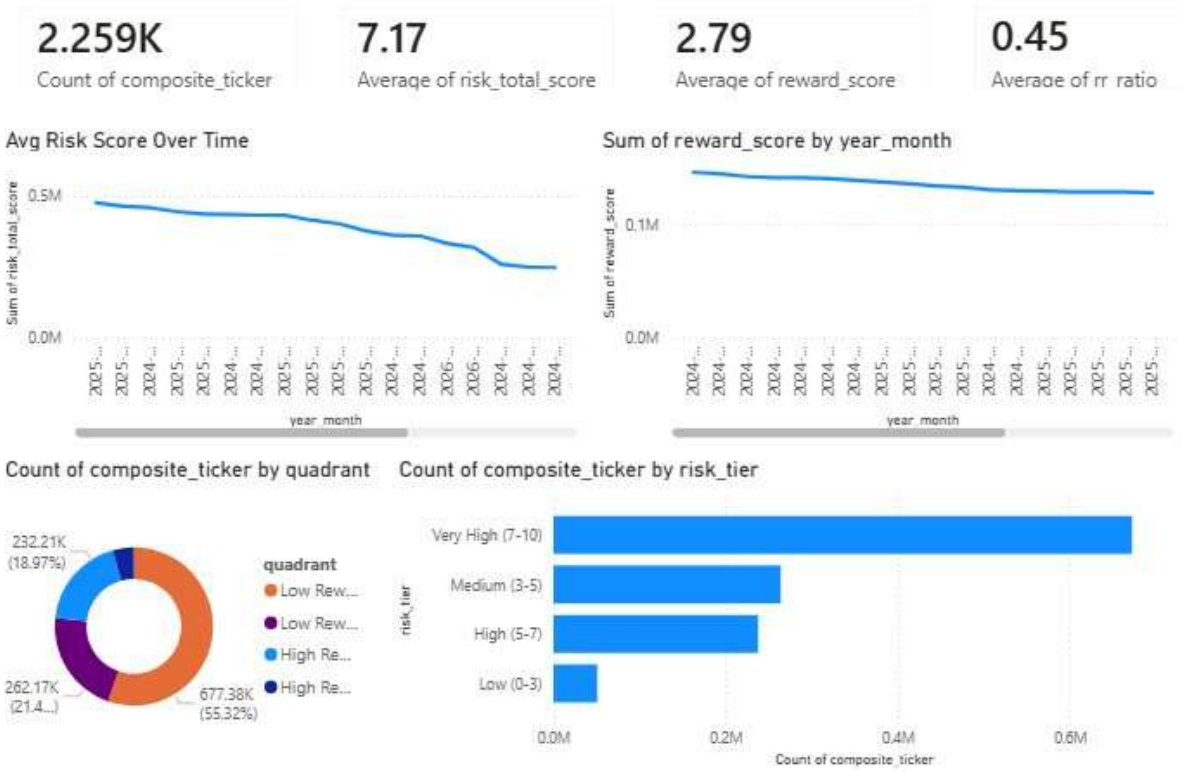
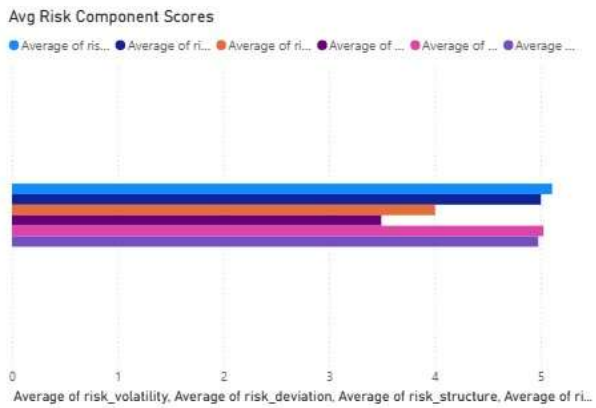


Exhibit 7 presents the Executive Overview page, displaying the four primary KPI metrics (2,259 ETFs, avg risk 7.17, avg reward 2.79, avg RR ratio 0.45) alongside time-series trend lines for both risk and reward scores over the 2024-2026 period. The donut chart confirms the quadrant distribution, with the Low Reward / High Risk segment accounting for 55.32% of all observations. The risk tier bar chart visualizes the heavy skew toward Very High risk, with that tier representing the largest segment by a substantial margin.

Exhibit 8: Power BI Dashboard - Risk Deep Dive (Page 2)



composite_ticker	Average of risk_total_score	risk_tier
RSX	9.95	Very High (7-10)
DAG	9.84	Very High (7-10)
LBJ	9.80	Very High (7-10)
SOXL	9.71	Very High (7-10)
UMDD	9.64	Very High (7-10)
RUSL	9.63	Very High (7-10)
TMF	9.63	Very High (7-10)
FAS	9.62	Very High (7-10)
UGLD	9.59	Very High (7-10)
FWDI	9.58	Very High (7-10)
UDOW	9.57	Very High (7-10)
EDC	9.56	Very High (7-10)
URTY	9.56	Very High (7-10)
TNA	9.55	Very High (7-10)
NUGT	9.55	Very High (7-10)
USLV	9.52	Very High (7-10)
FONE	9.49	Very High (7-10)
Total	**7.17**	

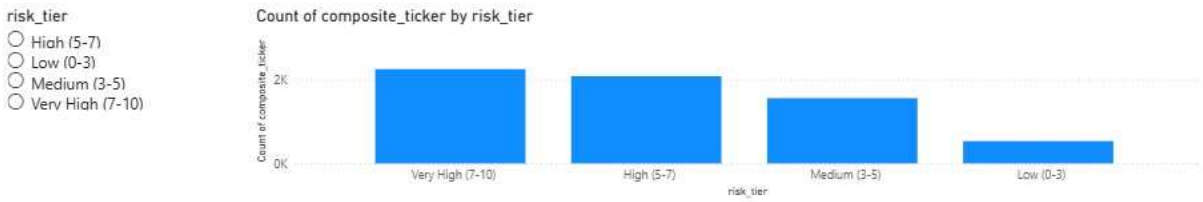


Exhibit 8 presents the Risk Deep Dive page, showing average risk component scores across the six ETFG dimensions. Volatility clearly dominates as the longest bar, consistent with its mean score of 5.11. The Top 20 Riskiest ETFs table on the right identifies funds including RSX (9.95), DAG (9.84), and SOXL (9.71) as the highest-risk products in the universe - all classified as Very High (7-10). The risk tier slicer enables dynamic filtering so users can isolate component behavior within each tier.

Exhibit 9: Power BI Dashboard - Risk-Reward Quadrant Analysis (Page 3)

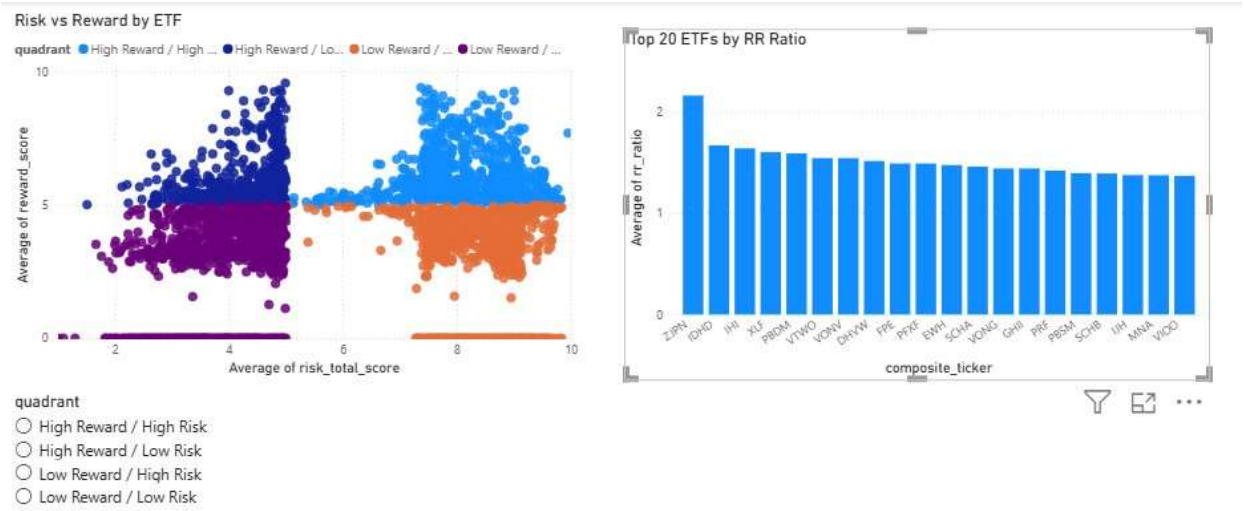
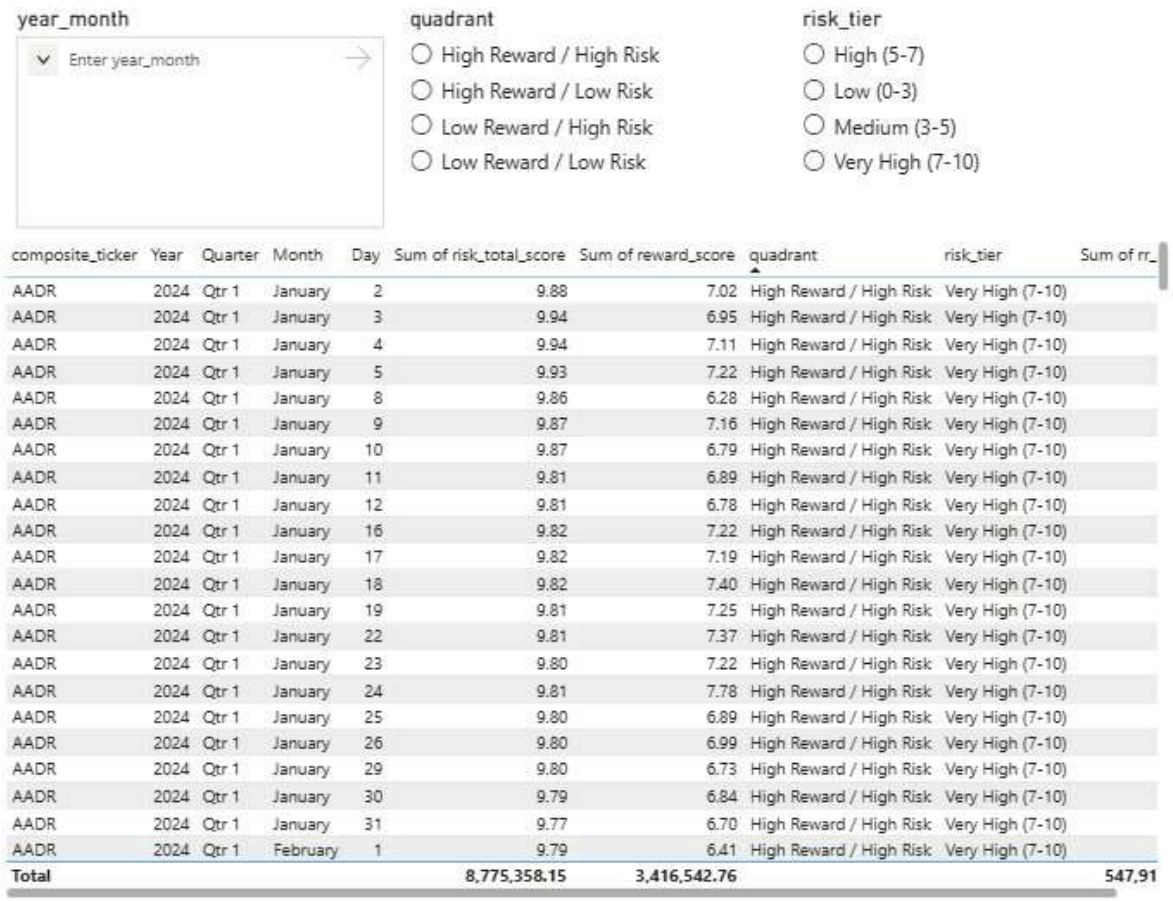


Exhibit 9 presents the Quadrant Analysis page, featuring the scatter plot of average risk score (X-axis) against average reward score (Y-axis) for each ETF, color-coded by quadrant. The visual confirms the near-zero correlation finding (r = 0.0082): there is no discernible upward slope in the scatter, with dots distributed broadly across all risk levels regardless of reward score. The Top 20 ETFs by RR Ratio bar chart on the right identifies ZJPN as the clear leader at 2.16, followed by IDHD, IHI, and XLF. The quadrant slicer allows users to isolate any single quadrant for detailed examination.

Exhibit 10: Power BI Dashboard - ETF Screener (Page 4)



composite_ticker	Year	Quarter	Month	Day	Sum of risk_total_score	Sum of reward_score	quadrant	risk_tier	Sum of rr_
AADR	2024	Qtr 1	January	2	9.88	7.02	High Reward / High Risk	Very High (7-10)	
AADR	2024	Qtr 1	January	3	9.94	6.95	High Reward / High Risk	Very High (7-10)	
AADR	2024	Qtr 1	January	4	9.94	7.11	High Reward / High Risk	Very High (7-10)	
AADR	2024	Qtr 1	January	5	9.93	7.22	High Reward / High Risk	Very High (7-10)	
AADR	2024	Qtr 1	January	8	9.86	6.28	High Reward / High Risk	Very High (7-10)	
AADR	2024	Qtr 1	January	9	9.87	7.16	High Reward / High Risk	Very High (7-10)	
AADR	2024	Qtr 1	January	10	9.87	6.79	High Reward / High Risk	Very High (7-10)	
AADR	2024	Qtr 1	January	11	9.81	6.89	High Reward / High Risk	Very High (7-10)	
AADR	2024	Qtr 1	January	12	9.81	6.78	High Reward / High Risk	Very High (7-10)	
AADR	2024	Qtr 1	January	16	9.82	7.22	High Reward / High Risk	Very High (7-10)	
AADR	2024	Qtr 1	January	17	9.82	7.19	High Reward / High Risk	Very High (7-10)	
AADR	2024	Qtr 1	January	18	9.82	7.40	High Reward / High Risk	Very High (7-10)	
AADR	2024	Qtr 1	January	19	9.81	7.25	High Reward / High Risk	Very High (7-10)	
AADR	2024	Qtr 1	January	22	9.81	7.37	High Reward / High Risk	Very High (7-10)	
AADR	2024	Qtr 1	January	23	9.80	7.22	High Reward / High Risk	Very High (7-10)	
AADR	2024	Qtr 1	January	24	9.81	7.78	High Reward / High Risk	Very High (7-10)	
AADR	2024	Qtr 1	January	25	9.80	6.89	High Reward / High Risk	Very High (7-10)	
AADR	2024	Qtr 1	January	26	9.80	6.99	High Reward / High Risk	Very High (7-10)	
AADR	2024	Qtr 1	January	29	9.80	6.73	High Reward / High Risk	Very High (7-10)	
AADR	2024	Qtr 1	January	30	9.79	6.84	High Reward / High Risk	Very High (7-10)	
AADR	2024	Qtr 1	January	31	9.77	6.70	High Reward / High Risk	Very High (7-10)	
AADR	2024	Qtr 1	February	1	9.79	6.41	High Reward / High Risk	Very High (7-10)	
Total					**8,775,358.15**	**3,416,542.76**			**547,91**

Exhibit 10 presents the ETF Screener page, providing a fully interactive table of all ETF observations filterable by year-month, quadrant, and risk tier. This page serves as the primary tool for practitioners seeking to identify specific ETFs meeting defined risk-reward criteria. Users can filter to the High Reward / Low Risk quadrant and Low (0-3) risk tier simultaneously to generate a shortlist of optimal ETF candidates, supporting the RR ratio screening recommendation outlined in Section 5.3.

5. Final Conclusion and Call to Action

5.1 Summary of Findings

This analysis of 1,224,378 daily ETF observations across 2,259 funds from January 2024 to March 2026 yields five principal conclusions. First, the ETF universe is characterized by a systematic risk-reward imbalance: the mean risk score of 7.17 far exceeds the mean reward score of 2.79, producing a mean RR ratio of only 0.39. Second, risk and reward are essentially uncorrelated (r =

0.0082) - bearing more ETF risk does not translate into higher reward scores, directly challenging efficient market assumptions. Third, the majority of ETFs occupy the worst possible quadrant: 55.3% of all ETF-day observations fall in the Low Reward / High Risk category, while only 4.3% occupy the optimal High Reward / Low Risk quadrant. Fourth, reward is exceptionally scarce - 25% of ETF-day observations carry a reward score of exactly zero, and the top quartile of reward scores begins only at 4.82. Fifth, the top-performing ETFs by RR ratio (led by ZJPN at 2.16, XLF at 1.60, and VTWO at 1.54) share common characteristics of defined factor tilts, liquid mandates, and transparent index methodologies.

5.2 Significance

These findings carry significant implications for both retail and institutional investors. The pervasive risk-reward imbalance documented here suggests that investors who select ETFs without systematic risk-reward evaluation are likely to hold portfolios skewed toward high-risk, low-reward instruments. The ETFG framework provides a practical tool for addressing this problem, enabling investors to screen the 2,259-ETF universe for funds that genuinely compensate for the risks they carry.

From a market structure perspective, the prevalence of Very High risk scores raises questions about the risk disclosure practices of ETF sponsors and the adequacy of regulatory frameworks in communicating multi-dimensional risk to retail investors. The finding that one quarter of the ETF universe carries zero reward - while simultaneously carrying non-trivial risk - suggests a category of products that may not serve retail investor interests.

5.3 Recommendations

Based on this analysis, the following recommendations are offered for investors, portfolio managers, and further researchers:

- Adopt RR Ratio Screening: Prioritize ETFs with RR ratios above 1.0 as a baseline screen. This threshold eliminates the majority of the poor-value universe while retaining a manageable selection of genuinely risk-efficient funds.
- Systematically Avoid the Low Reward / High Risk Quadrant: Leveraged, inverse, and single-country emerging market ETFs consistently populate the worst quadrant.

Systematic exclusion of these products from core allocations is supported by the distributional evidence.

- Monitor Liquidity Risk Dynamically: Given the time-varying nature of liquidity risk scores - which spike during stress precisely when liquidity is most needed - quarterly review of ETF liquidity profiles is recommended for institutional holders.
- Favor Simple, Large, Cheap ETFs: The empirical evidence consistently favors broad-mandate ETFs with large AUM and low expense ratios. These funds dominate the High Reward / Low Risk quadrant and offer the best risk-adjusted profiles.
- Extend to Fund Flow Analysis: Future research should incorporate ETFG fund flow data to test whether reward scores predict subsequent capital inflows, potentially enabling a momentum-based ETF selection strategy grounded in ETFG scoring.

5.4 Limitations and Future Research

This study has several limitations worth acknowledging. First, the analysis relies on the ETFG proprietary scoring methodology, the precise algorithmic details of which are not publicly disclosed. Second, the quadrant analysis uses a fixed midpoint of 5.0 for both risk and reward dimensions, which may not reflect the optimal threshold for all investment contexts. Third, the study does not control for broader market conditions that may have influenced score dynamics over the 2024-2026 period.

Future research could extend this analysis in several directions: incorporating the additional quant scoring variables available in the ETFG dataset (technical, sentiment, fundamental, and global sub-scores); applying machine learning classification to predict quadrant membership from fund characteristics; and extending the panel backward to the 2020-2022 market cycle to examine risk-reward dynamics across different market regimes.

References

Agapova, A. (2011). Conventional mutual index funds versus exchange-traded funds. Journal of Financial Markets, 14(2), 323–343.

Ben-David, I., Franzoni, F., & Moussawi, R. (2018). Do ETFs increase volatility? Journal of Finance, 73(6), 2471–2535.

BlackRock. (2024). Global ETF research: 2024 midyear outlook. BlackRock Investment Institute.

Exchange Traded Funds Global (ETFG). (2024). Red Diamond and Green Diamond scoring methodology. ETFG Analytics Documentation. Retrieved from Wharton Research Data Services.

Israeli, D., Lee, C. M. C., & Sridharan, S. A. (2017). Is there a dark side to exchange traded funds? An information perspective. Review of Accounting Studies, 22(3), 1048–1083.

Rompotis, G. G. (2009). Interfamily competition on index tracking: The case of the Vanguard ETFs and index funds. Journal of Asset Management, 10(4), 263–278.

Wharton Research Data Services (WRDS). (2024). ETFG Analytics database documentation. University of Pennsylvania. https://wrds-www.wharton.upenn.edu